Modal Theory v12: Self-Consistent Geometric–Dynamical Closure and the 255° Attractor

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Abstract

Modal Theory v12 presents a flat-space two-scalar framework intended to unify the Standard Model, gravity, and cosmology through a single geometric–dynamical closure structure. The theory is built from two real scalar fields Φ_1 and Φ_2 interacting through a gravitational-strength coupling $g_{\mathrm{mode}} = 4\pi G$, with the relative phase θ embedded on the equator of S^2 and constrained by a Z_3 orbifold together with a Z_2 chiral bias.

The S^2 embedding fixes the harmonic grammar of the phase sector. The symmetric potential is identified as the lowest Fourier harmonic compatible with the Z_3 orbifold, giving the baseline term $-\cos(3\theta)$. Projection onto the 60° chiral axis fixes the $\sqrt{3}$ coefficient in the bias sector. The central result of v12 is a geometric derivation of the 15° offset from the 240° baseline minimum to the 255° attractor. A hexagonal reciprocal-space baseline has a 60° fundamental symmetry sector. The maximal inequivalent chiral twist is therefore 30°. Superposing the original and 30°-rotated reciprocal lattices yields a 12-direction interference superstructure with a 30° fundamental domain. Least-frustration relaxation to the bisector of that domain produces a precessional offset of $15° = \pi/12$. Applied to the 240° baseline, this yields the attractor $255° = 17\pi/12$.

The microscopic interference structure is then integrated out into a coarse-grained infrared envelope, giving a periodic low-energy phase potential centered on the derived attractor and avoiding double-counting of the ultraviolet geometric generators. In this way, the 255° phase appears as the natural candidate output of the full geometric–dynamical closure.

1 Introduction

Modal Theory begins from the assumption that the vacuum is described by two coupled real scalar fields and that the observable structure of physics emerges from the phase geometry of their interaction. The aim is to replace ad hoc parameter-setting by a self-consistent geometric–dynamical closure.

The central phase variable θ is embedded on the equator of S^2, subject to a Z_3 orbifold symmetry and a Z_2 chiral bias. Earlier versions of the theory established the threefold baseline structure, the geometric form of the bias term, and the existence of a preferred displaced basin near 255°. The missing step was to show why the shift from 240° to 255° is not arbitrary.

The purpose of v12 is to make that step explicit. The 15° offset is derived here as a geometric precession arising from the interaction of:

- the Z_3 structural baseline,
- the Z_2 chiral twist,
- the resulting 12-direction interference superstructure,
- and the relaxation of the phase vector to the bisector of the fundamental interference domain.

The result is a closed geometric sequence:

$$Z_3 \;\to\; Z_2 \;\to\; Z_{12} \;\to\; \frac{\pi}{12} \;\to\; \frac{17\pi}{12}.$$

2 Phase Topology and Orbifold Structure

The phase θ lives on the circle S^1, but physical admissibility is constrained by a Z_3 orbifold identification:

$$\theta \sim \theta + \frac{2\pi}{3}.$$

The fundamental domain is therefore

$$\theta \in \left[0, \frac{2\pi}{3}\right),$$

or equivalently $[0, 120°)$.

This produces three classical sectors corresponding to the anchor coordinates

$$\theta \in \left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}.$$

3 Symmetric Z_3 Potential from Orbifold Invariance

The symmetric phase potential must be invariant under the orbifold identification

$$\theta \mapsto \theta + \frac{2\pi}{3}.$$

Therefore any admissible Fourier mode must satisfy

$$\cos\big(m(\theta + 2\pi/3)\big) = \cos(m\theta),$$

which requires m to be a multiple of 3.

The lowest nontrivial harmonic compatible with this symmetry is therefore $m = 3$, giving

$$V_{Z_3}(\theta) = -\lambda_3 \cos(3\theta). \tag{1}$$

This is not an arbitrary choice of harmonic. It is the lowest Fourier mode allowed by the Z_3 orbifold.

The minima occur at

$$\theta = 0,\ \frac{2\pi}{3},\ \frac{4\pi}{3},$$

corresponding to 0°, 120°, and 240°.

4 Chiral Bias from the S^2 Embedding

The phase θ is embedded as the azimuthal coordinate on the equator of the unit sphere S^2:

$$\mathbf{n}(\vartheta, \phi) = (\sin\vartheta\cos\phi,\ \sin\vartheta\sin\phi,\ \cos\vartheta), \qquad \vartheta = \frac{\pi}{2} \implies \phi \equiv \theta.$$

A chiral axis at 60° on the equator is represented by the unit vector

$$\mathbf{e}_{60} = \left(\cos\frac{\pi}{3}, \sin\frac{\pi}{3}\right) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

Projecting the phase vector

$$\mathbf{n}_{\mathrm{eq}}(\theta) = (\cos\theta, \sin\theta)$$

onto this axis gives

$$\mathbf{n}_{\mathrm{eq}}(\theta) \cdot \mathbf{e}_{60} = \frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta.$$

Up to an overall factor of 2, this yields the geometric chiral combination

$$\cos\theta + \sqrt{3}\sin\theta.$$

Accordingly, the chiral bias sector is written as

$$V_{\mathrm{bias}}(\theta) = \chi\, k\big(\cos\theta + \sqrt{3}\sin\theta\big), \qquad \chi = \pm 1, \quad k > 0. \tag{2}$$

This separates the magnitude k from the branch choice χ. In the branch relevant to the 255° attractor, we choose

$$\chi = +1,$$

so that

$$V_{\mathrm{bias}}(\theta) = +k\big(\cos\theta + \sqrt{3}\sin\theta\big) = +2k\cos\left(\theta - \frac{\pi}{3}\right). \tag{3}$$

This term is minimized when

$$\cos\left(\theta - \frac{\pi}{3}\right) = -1,$$

which gives

$$\theta - \frac{\pi}{3} = \pi \qquad \Rightarrow \qquad \theta = \frac{4\pi}{3} = 240^\circ.$$

Thus the chiral sector selects the 240° branch of the Z_3 baseline.

5 Hexagonal Reciprocal Baseline from Z_3 Directions and Real-Mode Pairing

The underlying real-space structure is taken to be hexagonal. In reciprocal space, the physically relevant baseline consists of six directions separated by 60°. This can be understood as the real-field completion of the Z_3 directional structure.

A Z_3 set provides three directions,

$$0^\circ,\ 120^\circ,\ 240^\circ.$$

For a real field, each mode must be accompanied by its opposite wavevector $\mathbf{q} \leftrightarrow -\mathbf{q}$. The result is a six-direction reciprocal star:

$$0^\circ,\ 60^\circ,\ 120^\circ,\ 180^\circ,\ 240^\circ,\ 300^\circ.$$

Equivalently, the reciprocal vectors are

$$\mathbf{q}_n = |\mathbf{q}| \left(\cos \frac{n\pi}{3}, \sin \frac{n\pi}{3} \right), \qquad n = 0, 1, 2, 3, 4, 5. \tag{4}$$

Thus the Z_6 reciprocal baseline is not an extra assumption independent of the phase theory. It is the natural real-mode completion of the Z_3 structural directions.

6 Derivation of the 255° Attractor via Moiré Interference

The reciprocal-space baseline has a fundamental symmetry sector of

$$60^\circ = \frac{\pi}{3}.$$

The Z_2 chiral bias is represented as a continuous twist operator acting on the reciprocal lattice:

$$\mathbf{q}'_n = R(\alpha)\, \mathbf{q}_n. \tag{5}$$

A twist of $\alpha = 0^\circ$ is trivial, while a twist of $\alpha = 60^\circ$ is symmetry-equivalent to the next hexagonal slot. The maximal inequivalent twist is therefore the midpoint of the symmetry sector:

$$\alpha_{\max} = \frac{1}{2} \times 60^\circ = 30^\circ = \frac{\pi}{6}. \tag{6}$$

Superposing the original reciprocal lattice with its 30°-rotated copy produces a 12-direction interference superstructure. The original directions are

$$0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ,$$

while the rotated directions are

$$30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ, 330^\circ.$$

Their union is a set of twelve equally spaced directions. Equivalently,

$$\mathbf{Q}_m = |\mathbf{q}| \left(\cos \frac{m\pi}{6}, \sin \frac{m\pi}{6} \right), \qquad m = 0, 1, \dots, 11. \tag{7}$$

The fundamental angular domain of this interference structure is therefore

$$\Lambda_{12} = \frac{2\pi}{12} = \frac{\pi}{6} = 30^\circ. \tag{8}$$

7 Least-Frustration Bisector and the 15° Offset

Within a 30° interference domain, the boundaries correspond to maximal mismatch between the original and twisted structures. The least-frustrated phase position lies at the bisector of the domain.

A minimal toy frustration energy that captures this symmetry is

$$E(\delta) = A\delta^2 + A(\Lambda_{12} - \delta)^2, \qquad A > 0, \tag{9}$$

where δ measures the displacement from one boundary. Minimising gives

$$\frac{dE}{d\delta} = 2A\delta - 2A(\Lambda_{12} - \delta) = 0,$$

hence

$$\delta = \frac{\Lambda_{12}}{2}. \tag{10}$$

Therefore the precessional offset is

$$\theta_{\text{offset}} = \frac{1}{2}\Lambda_{12} = \frac{1}{2} \cdot \frac{\pi}{6} = \frac{\pi}{12} = 15°. \tag{11}$$

Applying this offset to the selected baseline minimum at 240° yields

$$\theta_* = \theta_{\text{base}} + \theta_{\text{offset}} = \frac{4\pi}{3} + \frac{\pi}{12} = \frac{16\pi}{12} + \frac{\pi}{12} = \frac{17\pi}{12}. \tag{12}$$

Hence

$$\theta_* = 255°. \tag{13}$$

The 15° shift is therefore not an arbitrary correction. It is the geometric consequence of maximal inequivalent chiral twisting of the hexagonal reciprocal baseline, followed by least-frustration relaxation to the bisector of the resulting 30° interference cell.

8 Coarse-Grained Infrared Envelope

At this point the ultraviolet geometric ingredients have already done their work:

- the Z_3 sector generated the baseline anchor,
- the Z_2 sector generated the chiral twist,
- the interference construction produced the 15° offset,
- and the attractor $\theta_* = 17\pi/12$ has been derived.

The correct low-energy description is therefore not obtained by adding the ultraviolet generators again as independent infrared forces. Doing so would double-count the same geometric bias.

Instead, the microscopic 12-direction interference structure is integrated out into a coarse-grained infrared envelope centered on the derived attractor:

$$V_{\mathrm{IR}}(\theta) = E_0 - \Lambda_1 \cos(\theta - \theta_*) - \Lambda_2 \cos\big(2(\theta - \theta_*)\big) - \cdots . \tag{14}$$

To leading order, the low-energy phase potential is therefore

$$V_{\mathrm{IR}}(\theta) \approx E_0 - \Lambda_1 \cos(\theta - \theta_*). \tag{15}$$

With

$$\theta_* = \frac{17\pi}{12},$$

this becomes

$$V_{\mathrm{IR}}(\theta) \approx E_0 - \Lambda_1 \cos\left(\theta - \frac{17\pi}{12}\right). \tag{16}$$

This is the correct periodic locking form on S^1. The 255° phase is not inserted by hand here; it is the center of the coarse-grained envelope produced by the microscopic geometry.

9 Stationarity and Local Stability

The infrared stationarity condition is

$$\frac{dV_{\mathrm{IR}}}{d\theta} = 0, \tag{17}$$

and to leading order gives

$$\Lambda_1 \sin(\theta - \theta_*) = 0. \tag{18}$$

Thus the stationary points occur at

$$\theta = \theta_* + n\pi.$$

Local stability requires

$$\frac{d^2 V_{\mathrm{IR}}}{d\theta^2} > 0. \tag{19}$$

At leading order,

$$\frac{d^2 V_{\mathrm{IR}}}{d\theta^2} = \Lambda_1 \cos(\theta - \theta_*), \tag{20}$$

so the attractor is locally stable at

$$\theta = \theta_* = \frac{17\pi}{12} = 255^\circ$$

provided

$$\Lambda_1 > 0.$$

10 Interpretation on S^2 and S^1

The S^2 embedding fixes the angular grammar of the theory: the Z_3 harmonic, the chiral projection, and the geometric meaning of the phase coordinate. The reciprocal-space construction then determines the precessional shift from 240° to 255°. Finally, the coarse-grained envelope gives the correct infrared locking on S^1.

The logic is therefore

$$\begin{aligned} S^2 \text{ geometry } &\to Z_3 \text{ baseline } \to Z_2 \text{ chiral twist} \\ &\to 12\text{-direction interference } \to 15^\circ \text{ precession} \\ &\to 255^\circ \text{ attractor } \to S^1 \text{ periodic IR lock.} \end{aligned}$$

11 RG Flow and Closure

At low energy, the phase evolution may be interpreted as a gradient flow of the infrared envelope:

$$\frac{d\theta}{d\ell} = -\frac{dV_{\mathrm{IR}}}{d\theta}. \tag{21}$$

To leading order,

$$\frac{d\theta}{d\ell} = -\Lambda_1 \sin(\theta - \theta_*), \tag{22}$$

so the closure-selected fixed point is

$$\theta^* = \theta_* = \frac{17\pi}{12} = 255^\circ.$$

In this way, the RG flow, the periodic locking sector, and the geometric derivation all converge on the same attractor.

12 Full Geometric–Dynamical Closure

The final closure sequence is now explicit:

1. The Z_3 orbifold fixes the baseline harmonic $-\cos(3\theta)$ and the anchor minima.
2. The S^2 embedding fixes the geometric form of the chiral bias and the $\sqrt{3}$ coefficient.
3. The 240° branch is selected by the positive-sign chiral sector.
4. Real-mode pairing of the Z_3 directions generates the hexagonal reciprocal baseline.
5. The maximal inequivalent Z_2 twist is 30°.
6. Superposition of the original and twisted reciprocal lattices generates a 12-direction interference superstructure.

7. Least-frustration relaxation to the bisector of the 30° interference domain gives the precessional offset $15^\circ = \pi/12$.

8. Applying that offset to the 240° baseline minimum yields

$$240^\circ + 15^\circ = 255^\circ.$$

9. Integrating out the microscopic interference structure produces a periodic infrared envelope centered on the derived attractor.

In this way, the 255° phase is not a phenomenological insertion. It is the geometric consequence of the twisted hexagonal baseline together with the correct coarse-grained periodic locking on the circle.

All downstream predictions inherit this closure structure.